

Semiconductor Process Yield & Anomaly Triage

Rare-event ranking under small-n, high-dimensional manufacturing constraints

OPERATING QUESTION

If engineers can review only 10% of records, how many failures concentrate at the top of the ranked queue?

0.206

AVERAGE PRECISION

vs 0.054 holdout prevalence

5 of 17

FAILURES CAPTURED

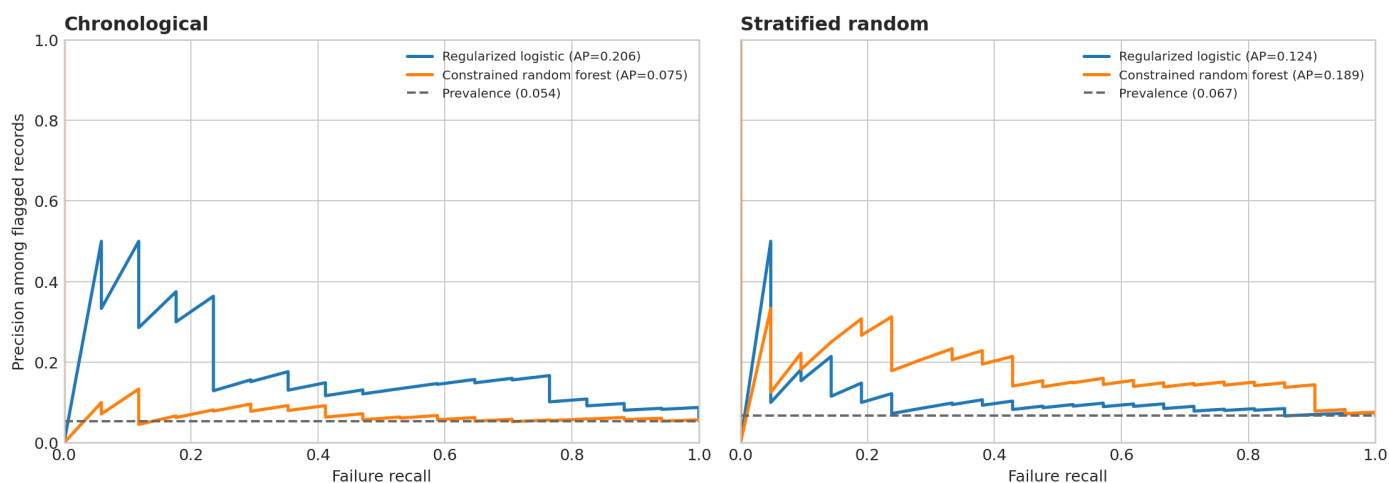
top 32 of 314 records

2.9x

LIFT AT 10%

29% recall; 16% precision

Precision-recall performance changes with validation design



METHOD

Primary split: first 80% train, last 20% test by timestamp.
Sensitivity: stratified random 80/20 split.
Models: class-weighted regularized logistic regression and one constrained random forest.
Feature filtering, median imputation, missingness indicators, and scaling are learned on training data only.

INTERPRETATION BOUNDARY

Anonymous features can prioritize engineering review and generate hypotheses. They cannot be assigned physical meanings or called root causes. Model scores are ranking signals, not calibrated failure probabilities. The simulated shift exercise is a monitoring illustration, not evidence of a historical fab event.

CENTRAL FINDING

The model ranking reverses across split designs: logistic regression leads on the chronological holdout, while the forest leads under random splitting. Validation design therefore changes the substantive conclusion. The chronological logistic AP bootstrap interval is 0.108 to 0.424.

NEXT EXPERIMENT

Validate on newer forward cohorts; add tool, chamber, recipe, product, maintenance, and genealogy context; define the operating threshold from review capacity and false-alarm cost; then capture reviewer feedback prospectively.